

EMERGING ETHICS: TECHNOLOGY, AI AND ENVIRONMENT

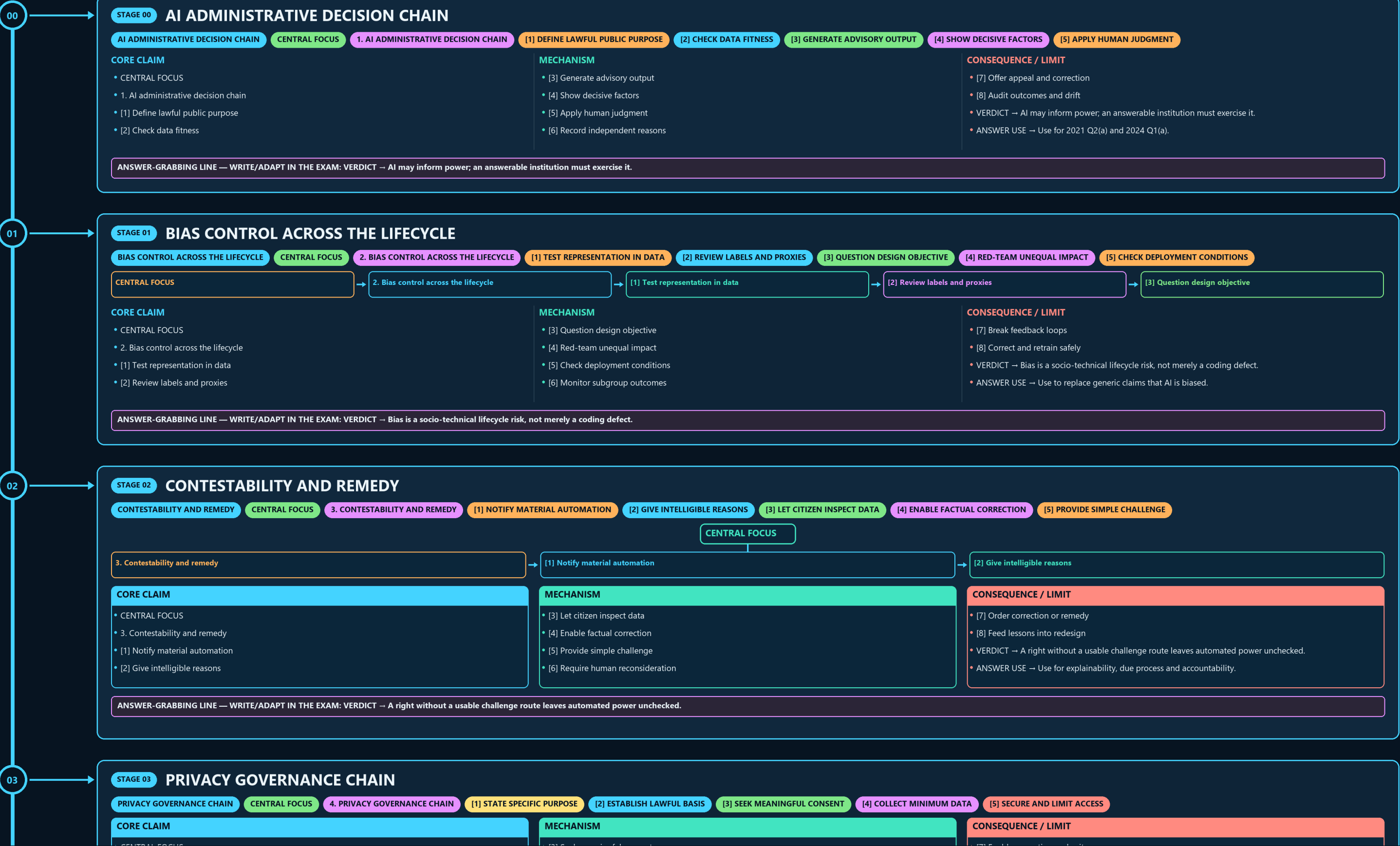
AI ADMINISTRATIVE DECISION CHAIN → BIAS CONTROL ACROSS THE → CONTESTABILITY AND REMEDY → PRIVACY GOVERNANCE CHAIN → PUTTASWAMY PROPORTIONALITY TEST → SOCIAL-MEDIA ETHICS MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g1 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



- [8] Feed lessons into redesign
- VERDICT → A right without a usable challenge route leaves automated power unchecked.
- ANSWER USE → Use for explainability, due process and accountability.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → A right without a usable challenge route leaves automated power unchecked.

STAGE 03 PRIVACY GOVERNANCE CHAIN

[5] SECURE AND LIMIT ACCESS

CONSEQUENCE / LIMIT

- [7] Enable correction and exit
- [8] Remedy breach and misuse
- VERDICT → Privacy protects dignity and autonomy throughout the data lifecycle.
- ANSWER USE → Use for consent, purpose and minimisation answers.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Privacy protects dignity and autonomy throughout the data lifecycle.

STAGE 04 PUTTASWAMY PROPORTIONALITY TEST

[5] TEST LEAST-RESTRICTIVE MEANS

CONSEQUENCE / LIMIT

- [7] Add procedural safeguards
- [8] Keep review and remedy open
- VERDICT — A useful aim does not by itself justify a privacy restriction.
- ANSWER USE — Use as the constitutional spine for State data use.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → A useful aim does not by itself justify a privacy restriction.

STAGE 05 SOCIAL-MEDIA ETHICS MAP

[5] CYBERBULLYING AND DIGNITY

CENTRAL FOCUS

→ [2] Misinformation and deepfakes

CONSEQUENCE / LIMIT

- [7] Moderation reasons and appeal
- [8] Verification and proportionate law
- VERDICT → Protect open debate while making hidden manipulation and preventable harm answerable.
- ANSWER USE → Use for 2023 Q12 and 2025 Q1(a).

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Protect open debate while making hidden manipulation and preventable harm answerable.

STAGE 06 AI ENVIRONMENTAL FOOTPRINT

[5] MINERAL AND SUPPLY-CHAIN HARM

→ [3] Cooling and power water

CONSEQUENCE / LIMIT

- [7] Absolute-impact disclosure
- [8] Budgets, efficiency and repair

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Protect open debate while making hidden manipulation and preventable harm answerable.

06

STAGE 06 AI ENVIRONMENTAL FOOTPRINT

AI ENVIRONMENTAL FOOTPRINT CENTRAL FOCUS 7. AI ENVIRONMENTAL FOOTPRINT [1] TRAINING ELECTRICITY DEMAND [2] REPEATED INFERENCE DEMAND [3] COOLING AND POWER WATER [4] HARDWARE EMBODIED IMPACTS [5] MINERAL AND SUPPLY-CHAIN HARM

CENTRAL FOCUS → 7. AI environmental footprint → [1] Training electricity demand → [2] Repeated inference demand → [3] Cooling and power water

CORE CLAIM

- CENTRAL FOCUS
- 7. AI environmental footprint
- [1] Training electricity demand
- [2] Repeated inference demand

MECHANISM

- [3] Cooling and power water
- [4] Hardware embodied impacts
- [5] Mineral and supply-chain harm
- [6] Replacement and e-waste

CONSEQUENCE / LIMIT

- [7] Absolute-impact disclosure
- [8] Budgets, efficiency and repair
- VERDICT → A renewable-power claim alone does not establish low-impact AI.
- ANSWER USE → Use for 2024 Q7 and lifecycle analysis.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → A renewable-power claim alone does not establish low-impact AI.

07

STAGE 07 PRECAUTION BEFORE SCALE-UP

PRECAUTION BEFORE SCALE-UP CENTRAL FOCUS 8. PRECAUTION BEFORE SCALE-UP [1] IDENTIFY SERIOUS POSSIBLE HARM [2] MAP UNCERTAINTY AND EXPOSURE [3] TEST REVERSIBLE PILOT [4] COMPARE SAFER ALTERNATIVES [5] APPLY PROPORTIONATE CONTROLS

CORE CLAIM

- CENTRAL FOCUS
- 8. Precaution before scale-up
- [1] Identify serious possible harm
- [2] Map uncertainty and exposure

MECHANISM

- [3] Test reversible pilot
- [4] Compare safer alternatives
- [5] Apply proportionate controls
- [6] Set stop and fallback rules

CONSEQUENCE / LIMIT

- [7] Monitor emerging evidence
- [8] Revise before wider deployment
- VERDICT → Rio and Vellore guide AI only by analogy, not as binding AI law.
- ANSWER USE → Use for anticipatory AI and environmental governance.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Rio and Vellore guide AI only by analogy, not as binding AI law.

08

STAGE 08 ENVIRONMENTAL JUSTICE TEST

ENVIRONMENTAL JUSTICE TEST CENTRAL FOCUS 9. ENVIRONMENTAL JUSTICE TEST [1] IDENTIFY ECOLOGICAL BENEFITS [2] IDENTIFY POLLUTION AND LOSS [3] MAP BURDENED COMMUNITIES [4] INCLUDE FUTURE GENERATIONS [5] SECURE MEANINGFUL PARTICIPATION

CORE CLAIM

- CENTRAL FOCUS
- 9. Environmental justice test
- [1] Identify ecological benefits
- [2] Identify pollution and loss

MECHANISM

- [3] Map burdened communities
- [4] Include future generations
- [5] Secure meaningful participation
- [6] Compare distributional options

CONSEQUENCE / LIMIT

- [7] Assign compensation and cleanup
- [8] Provide accessible remedy
- VERDICT → A green or welfare label cannot hide unequal burdens and silenced voices.
- ANSWER USE → Use for pollution, housing and siting cases.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → A green or welfare label cannot hide unequal burdens and silenced voices.

09

STAGE 09 SUSTAINABLE-DEVELOPMENT BRIDGE

SUSTAINABLE-DEVELOPMENT BRIDGE CENTRAL FOCUS 10. SUSTAINABLE-DEVELOPMENT BRIDGE [1] DEFINE PRESENT HUMAN NEED [2] VALUE ECOSYSTEM SERVICES [3] PROTECT FUTURE CAPABILITY [4] COMPARE LOWER-IMPACT ROUTES [5] INTERNALISE POLLUTION COSTS

CENTRAL FOCUS → 10. Sustainable-development bridge → [1] Define present human need → [2] Value ecosystem services → [3] Protect future capability

CORE CLAIM

- CENTRAL FOCUS
- 10. Sustainable-development bridge
- [1] Define present human need
- [2] Value ecosystem services

MECHANISM

- [3] Protect future capability
- [4] Compare lower-impact routes
- [5] Internalise pollution costs
- [6] Support a just transition

CONSEQUENCE / LIMIT

- [7] Monitor real outcomes
- [8] Revise if limits are breached
- VERDICT → Development and ecology are jointly constrained, not automatic overrides.
- ANSWER USE → Use for climate ethics and 2025 Q8.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Development and ecology are jointly constrained, not automatic overrides.

09

STAGE 09 SUSTAINABLE-DEVELOPMENT BRIDGE

SUSTAINABLE-DEVELOPMENT BRIDGE

CENTRAL FOCUS

10. SUSTAINABLE-DEVELOPMENT BRIDGE

[1] DEFINE PRESENT HUMAN NEED

[2] VALUE ECOSYSTEM SERVICES

[3] PROTECT FUTURE CAPABILITY

[4] COMPARE LOWER-IMPACT ROUTES

[5] INTERNALISE POLLUTION COSTS

CENTRAL FOCUS

10. Sustainable-development bridge

[1] Define present human need

[2] Value ecosystem services

[3] Protect future capability

CORE CLAIM

- CENTRAL FOCUS
- 10. Sustainable-development bridge
- [1] Define present human need
- [2] Value ecosystem services

MECHANISM

- [3] Protect future capability
- [4] Compare lower-impact routes
- [5] Internalise pollution costs
- [6] Support a just transition

CONSEQUENCE / LIMIT

- [7] Monitor real outcomes
- [8] Revise if limits are breached
- VERDICT → Development and ecology are jointly constrained, not automatic overrides.
- ANSWER USE → Use for climate ethics and 2025 Q8.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Development and ecology are jointly constrained, not automatic overrides.

10

STAGE 10 BORDER ENVIRONMENTAL CLEARANCE

BORDER ENVIRONMENTAL CLEARANCE

CENTRAL FOCUS

11. BORDER ENVIRONMENTAL CLEARANCE

[1] SPECIFY SECURITY OBJECTIVE

[2] MAP SENSITIVE ECOLOGY

[3] COMPARE ROUTE ALTERNATIVES

[4] USE LEAST-DAMAGING FEASIBLE DESIGN

[5] CONSULT ON NON-CLASSIFIED IMPACTS

CENTRAL FOCUS

11. Border environmental clearance

[1] Specify security objective

[2] Map sensitive ecology

CORE CLAIM

- CENTRAL FOCUS
- 11. Border environmental clearance
- [1] Specify security objective
- [2] Map sensitive ecology

MECHANISM

- [3] Compare route alternatives
- [4] Use least-damaging feasible design
- [5] Consult on non-classified impacts
- [6] Review secrets through cleared experts

CONSEQUENCE / LIMIT

- [7] Condition restoration and monitoring
- [8] Publish reasons within security limits
- VERDICT → Security warrants tailored protection, not automatic ecological exemption.
- ANSWER USE → Use for 2025 Q2(b).

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Security warrants tailored protection, not automatic ecological exemption.

11

STAGE 11 CLIMATE FAIRNESS ARCHITECTURE

CLIMATE FAIRNESS ARCHITECTURE

CENTRAL FOCUS

12. CLIMATE FAIRNESS ARCHITECTURE

[1] COMMON DUTY TO PROTECT CLIMATE

[2] HISTORICAL CONTRIBUTION MATTERS

[3] RESPECTIVE CAPABILITY MATTERS

[4] CURRENT EMISSIONS STILL MATTER

[5] CBDR-RC DIFFERENTIATES EFFORT

CENTRAL FOCUS

12. Climate fairness architecture

[1] Common duty to protect climate

[2] Historical contribution matters

[3] Respective capability matters

CORE CLAIM

- CENTRAL FOCUS
- 12. Climate fairness architecture
- [1] Common duty to protect climate
- [2] Historical contribution matters

MECHANISM

- [3] Respective capability matters
- [4] Current emissions still matter
- [5] CBDR-RC differentiates effort
- [6] Finance and technology enable action

CONSEQUENCE / LIMIT

- [7] Protect vulnerable transitions
- [8] Retain reviewable national duties
- VERDICT → Differentiated responsibility means fair action, not zero obligation.
- ANSWER USE → Use for 2024 Q2(b) and climate justice.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Differentiated responsibility means fair action, not zero obligation.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

1. ARCHITECTURE

2. CONCEPTS AND DISTINCTIONS

3. DETAILED TREATMENT

4. INSTITUTIONAL AND REFORM ARCHITECTURE

5. INDIAN APPLICATIONS AND BOUNDARY CASES

6. LIMITATIONS AND TRADE-OFFS

7. MUST-KNOW FACTS FOR ADVANCED PRELIMS

OPTIONAL ETHICAL DEPTH

- 1. Architecture
- 2. Concepts and distinctions
- 3. Detailed treatment
- 4. Institutional and reform architecture

ADVANCED LIMIT / TRADE-OFF

- 5. Indian applications and boundary cases
- 6. Limitations and trade-offs
- 7. Must-Know Facts for Advanced Prelims
- 8. Advanced Prelims traps

ADMINISTRATIVE APPLICATION

- 9. Current-anchor application
- 10. PYQ-based analytical application
- 11. Mains-ready framework
- 12. Probable questions

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.